

MCQ related to heme, liver, vitamins, minerals, nutrition and other.

Please take the 50 MCQ in 75 min and read the explanations!

The questions and explanations are provided by SGU SOM Biochemistry faculty.

Q#1	During your clinical rotations, one of your patients is a 25-year-old woman. Her eyes show “Kayser-Fleischer rings”. Which defect related to mineral metabolism leads to her condition? B. is correct: Toxic copper levels. This is a defect in biliary copper excretion due to a specific copper transporting ATPase deficiency in the liver. The copper is not released into the bile and not bound to ceruloplasmin.. Copper accumulates in this patient in the eye in form of characteristic rings. It leads also to copper accumulations in liver and brain. a. Copper deficiency [is incorrect, copper deficiency can lead to aneurisms] b. correct c. Iron deficiency [is incorrect, it would lead to microcytic anemia] d. Toxic iron levels [is incorrect, it would lead to hemochromatosis with iron accumulation in liver, pancreas and skin (cirrhosis, diabetes, grey, bronze skin)] e. Zinc deficiency [is incorrect, it would not lead to Kayser-Fleischer rings, zinc is needed as cofactor for more than 100 enzymes]
Q#2	Anemia can be grouped into normocytic, microcytic and macrocytic anemia. The deficiency of which of the following vitamins leads to microcytic anemia? D. is correct: vitamin B6. Vitamin B6 deficiency leads to PLP deficiency. Microcytic anemia is due to less hemoglobin synthesis in erythroid cells. PLP is needed for ALA synthase, and deficiency leads to less heme synthesis. If less heme is synthesized in erythroid cells, then also less globin chains are synthesized. This leads overall to less hemoglobin and microcytic anemia. a. vitamin B1 [is incorrect: TPP deficiency would lead to Wernicke-Korsakoff syndrome or Beriberi] b. vitamin B2 [is incorrect: FMN and FAD deficiency would lead to glossitis and cheilosis] c. vitamin B3 [is incorrect: NAD⁺ and NADPH deficiencies lead to pellagra] d. correct e. vitamin B12 [is incorrect: deficiency of vitamin B12 leads to macrocytic anemia (as does deficiency of THF), but not to microcytic anemia]
Q#3	The blood clotting cascade works via proteolytic activation of blood clotting factors. It is critical that the cascade is performed in a fast and efficient manner. For this reason, specific inactive blood clotting factors are γ-carboxylated in the liver (maturation of blood clotting factors), so that they can form calcium bridges to activated platelets during blood clotting. What is special about this process? The process of γ-carboxylation a. is correct: uses as coenzyme phyloquinone and menaquinone. (Vitamin K from plants and bacteria can be used.) b. is activated by warfarin in former stroke patients [is incorrect, warfarin reduces maturation of blood clotting factors in the liver] c. is activated during severe liver damage in chronic alcoholics [is incorrect, severe liver damage leads to less synthesis and less maturation of blood clotting factors, often vitamin K deficiency] d. uses biotin as coenzyme for the incorporation of carbon dioxide [is incorrect, biotin is a coenzyme for most carboxylations, but it is NOT used for γ-carboxylation, which needs vitamin K.]

	e. can totally be prevented by a diet low in phyloquinone [is incorrect, as intestinal bacteria form vitamin K independent of the low phyloquinone diet]
Q#4	An abnormality of copper metabolism can lead to severe health problems. Which of the following defects and diseases are related to low levels of ceruloplasmin and high toxic copper levels? E. is correct: Wilson's disease. This is due to a deficiency of biliary copper excretion and defective binding of copper to ceruloplasmin. Ceruloplasmin is much faster degraded when it does not bind copper. Copper levels increase and free copper is released into the blood. Copper accumulates in tissues like the eye (Kayser-Fleischer rings), brain and liver. a. Aneurisms [is incorrect, not high, but low levels of copper lead to aneurisms, lysyl oxidase needs copper as cofactor during elastin desmosine formation] b. Menke's syndrome [is incorrect, this is a genetic defect leading to low copper levels. It is characterized by "kinky" hair and cerebral dysfunction]. c. Rickets [is incorrect, this is due to deficiency of calcitriol and an abnormal calcium metabolism] d. Blue sclerae [is incorrect, this is mostly related to osteogenesis imperfecta leading to abnormal collagen] e. Correct
Q#5	The lipid soluble vitamins A and D lead to molecules that have steroid-hormone like actions in the nucleus. Vitamin A is part of the group of retinoids. Which retinoid is formed inside of cells in the cytosol and has in the nucleus steroid hormone-like action related to transcription? C is correct: retinoic acid. It is formed in target cells from retinol which is transported in blood to the cells bound to retinol binding protein (RBP). After uptake into the cell, retinol is used to form retinoic acid in the cytosol. Retinoic acid is then transported into the nucleus and has steroid-hormone like action related to transcription a. Retinol [is incorrect, retinol is transported in blood bound to RBP or it is stored in the liver as retinyl ester. Retinol does not have hormone-like action, retinoic acid does] b. Beta-carotene [is incorrect, it is taken up from the diet and cleaved to two retinals, leading to two retinols. It does not have hormone-like action] c. correct d. Retinal [is incorrect, retinal does not have hormone-like action in the nucleus, it can be formed from retinol in the retina] e. Cis-retinal [is incorrect, it is formed from cis-retinol in the retina and leads to trans retinal. It is used in the visual cycle when it is bound to opsin forming rhodopsin. Deficiency leads to night blindness as result of vitamin A deficiency]
Q#6	A two-year-old child is brought by his parents to the family physician. The child has soft, pliable bones and has difficulty to stand. The parents explain that the child did not like the taste of milk and cheese, and that they had kept him inside and away from sunlight as his skin was easily burnt by the sun. The

	physician diagnoses rickets and prescribes vitamin treatment with vitamin D. The physiological active form of vitamin D is calcitriol which has hormone-like action related to transcription. Where and how is calcitriol formed in the human body? In D. is correct: the kidney from 25-OH D. 25-OH D is hydroxylated in the kidney to 1,25-diOH D, also named calcitriol. This is the regulated step of the synthesis of the physiological active calcitriol. It is activated by low blood phosphorous and by parathyroid hormone, and it is inhibited by calcitriol itself (product inhibition). a. the skin from 7-dehydrocholesterol [is incorrect, vitamin D₃ (cholecalciferol) is formed and not calcitriol, which is 1,25-diOH D] b. the liver from dietary vitamin D [is incorrect, it is 25-OH D that is formed in the first hydroxylation step performed by the liver] c. the blood from 25-OH D [is incorrect, this molecule is not hydroxylated in the blood. It is transported in blood, and can be hydroxylated the second time in the kidney] d. correct e. intestinal cells from cholesterol [is incorrect, instead, the already active calcitriol enters the intestinal mucosal cell and acts in the nucleus leading to transcription of calcium binding protein.]
Q#7	The moderate deficiency of vitamin B6 leads to general tiredness, irritability and depression. Vitamin B6 is used to form the coenzyme pyridoxal phosphate which is needed by many enzymes. Which of the following enzymes needs pyridoxal phosphate as coenzyme? D. is correct: Aminotransferase. Aminotransferases are key enzymes for amino acid metabolism related to synthesis and also degradation of respective amino acids. This is especially important in the liver, in relation to the urea cycle. PLP deficiency leads to less activity of all aminotransferases. a. ALA dehydratase [is incorrect, this enzyme needs zinc. It is ALA synthase that needs PLP] b. Methionine synthase [is incorrect, this enzyme methylates homocysteine and needs methyl-THF and vitamin B12] c. Transketolase [is incorrect, this enzyme needs TPP as coenzyme formed from vitamin B1] d. correct e. Methylmalonyl-CoA mutase [is incorrect, the coenzyme is cobalamin. Vitamin B12 deficiency is characterized by methylmalonate in the urine.]
Q#8	Ascorbic acid is a vitamin (Vitamin C) in humans, as it cannot be formed in primates. A severe deficiency can lead to scurvy which is characterized by sore gums, loose teeth or loss of teeth, and fragile blood vessels. What is a major function of Vitamin C in humans? Vitamin C is needed for C. is correct: hydroxyprolyl formation during collagen synthesis. Hydroxyprolyl residues are introduced by prolyl-hydroxylase into pro-α- chains during collagen synthesis inside of fibroblasts. This enzyme needs vitamin C. These hydroxyproline residues are important to stabilize the triple-helix of collagen by many hydrogen bonds. a. dietary heme iron uptake [is incorrect, it is the uptake of inorganic iron which needs to be in the ferrous form. This is obtained by vitamin C or high proton concentration in the stomach]

	b. allysine formation for elastin and collagen cross-linking [<i>is incorrect</i>, allysine formation is catalyzed by lysyl oxidase which needs copper as cofactor] c. correct d. prevention of hemorrhagic disease of the newborn [<i>is incorrect</i>, it is vitamin K that is given to newborns to prevent hemorrhagic disease (insufficient γ-carboxylation of inactive blood clotting factors)] e. formation of tyrosine from phenylalanine [<i>is incorrect</i>, phenylalanine hydroxylase needs tetrahydrobiopterin as coenzyme. Note: do not mix it up with tetrahydrofolate! Tetrahydrobiopterin is not formed from a vitamin but synthesized starting with GTP in humans. It is also needed for tryptophan hydroxylase (serotonin synthesis) and tyrosine hydroxylase (DOPA, catecholamine synthesis)]
Q#9	Selenium is a trace mineral that is not needed as cofactor for many enzymes. Which of the following enzymes needs selenium as cofactor? D. is correct: glutathione peroxidase. This enzyme acts on hydrogen peroxide or on organic peroxides (ROOH) and needs two reduced glutathiones (2 GSH) for its reaction. It is also used for reversal of lipid peroxidation of polyunsaturated fatty acids (PUFA). a. Superoxide dismutase [<i>is incorrect</i>, in humans, SOD in cytosol needs zinc and copper, extracellular SOD needs copper and SOD in mitochondria needs manganese] b. Catalase [<i>is incorrect</i>, catalase needs iron for its heme group.] c. Glutathione reductase [<i>is incorrect</i>, it needs as cofactor FAD and uses NADPH] d. correct e. Nitric oxide synthase [<i>is incorrect</i>, it needs iron in heme, FMN, FAD, tetrahydrobiopterin and uses NADPH]
Q#10	The enzyme xanthine oxidase catalyzes the reactions from hypoxanthine to xanthine and also from xanthine to uric acid. Which of the following trace minerals is needed as cofactor in both reactions? b. is correct: molybdenum. This cofactor is not used by many enzymes. Xanthine oxidase has also iron-sulfur clusters. a. Chromium [<i>is incorrect</i>, it is needed for glucose tolerance factor] b. correct c. Manganese [<i>is incorrect</i>, it is cofactor for pyruvate carboxylase and mitochondrial SOD] d. Zinc [<i>is incorrect</i>, zinc is used for many enzymes, but not for xanthine oxidase] e. Copper [<i>is incorrect</i>, copper is used for ceruloplasmin, lysyl oxidase, tyrosinase, cytochrome c]
Q#11	Biotin is a vitamin grouped in the B-complex as energy-releasing vitamin. Biotin is covalently bound to a lysine residue of specific enzymes, and then it is often referred to as biocytin. Which of the following enzymes needs biotin as prosthetic group? A. is correct: acetyl CoA carboxylase. This enzyme forms malonyl CoA in cytosol for fatty acid de novo synthesis. Also, the carboxylations in mitochondria performed by pyruvate carboxylase (formation of oxaloacetate), and by propionyl-CoA carboxylase (formation of methylmalonyl-CoA) need biotin. a. correct b. Carbamoyl-phosphate synthase I [<i>is incorrect</i>, this carboxylation using carbon dioxide does not need biotin. It is special that the enzyme needs N-acetyl-glutamate as absolute activator. This is an indicator for high glutamate]

	and arginine levels and the need for the urea cycle to take place] c. Enzymes performing γ-carboxylations [is incorrect, this carboxylation using carbon dioxide does not need biotin. It needs vitamin K] d. Methylmalonyl-CoA mutase [is incorrect, biotin is not needed, but vitamin B12 is needed in mitochondria] e. Pyruvate dehydrogenase [is incorrect, biotin is not needed. This enzyme complex needs thiamine-PP, lipoic acid, CoA, FAD and NAD⁺.]
Q#12	Methionine contains in its structure a methyl-group. In the activated form of methionine as S-adenosyl-methionine (SAM) this methyl group is used by many enzymes for methylation of their substrates. Which of the following compounds is methylated using SAM? C. is correct: Norepinephrine is methylated to epinephrine using SAM. a. Homocysteine [is incorrect, methylation of homocysteine to methionine needs methyl-THF and vitamin B12] b. Dopamine [is incorrect, dopamine is not methylated, it is hydroxylated to norepinephrine. The enzyme dopamine hydroxylase needs copper and vitamin C] c. correct d. dUMP [is incorrect, thymidylate synthase needs methylene-THF] e. Tetrahydrofolate [is incorrect, methylene-THF is used to form methyl-THF, irreversible]
Q#13	The trace mineral manganese is not needed as cofactor for many enzymes. Which of the following enzymes needs manganese as cofactor? a. is correct: Pyruvate carboxylase. This enzyme forms oxaloacetate in mitochondria and needs acetyl-CoA as absolute activator and manganese as cofactor. This enzyme forms oxaloacetate as anaplerotic (refill) reaction for the TCA cycle (in this case acetyl CoA is formed from PDH) or it forms oxaloacetate during gluconeogenesis (in this case acetyl-CoA formed by β-oxidation). a. correct b. Cytosolic superoxide dismutase [is incorrect, cytosolic superoxide dismutase needs copper and zinc as cofactors, manganese is needed as cofactor for mitochondrial superoxide dismutase] c. Dopamine hydroxylase [is incorrect, this enzyme needs copper and vitamin C] d. Tyrosinase [is incorrect, this enzyme (melanin formation) needs copper] e. Carbonic anhydrase [is incorrect, this enzyme needs zinc as cofactor]
Q#14	A 38-year old male changes his occupation and the new employer insists on a health check-up. The physician notices an abnormal slate-grey, bronze color of his skin. The blood analysis shows elevated levels of ALT, AST and ALP. The blood glucose level is 200 mg/dL. The physician suspects hereditary hemochromatosis and she orders additional blood tests. Which abnormalities at the biochemical level are found in this patient with hemochromatosis? a. is correct: High serum ferritin levels. Ferritin and hemosiderin are used to store iron inside of cells. In this 38-year old patients with hemochromatosis, ferritin leaks into the blood and leads to high serum ferritin levels. a. correct b. Deficiency of ceruloplasmin (ferroxidase)

	[is incorrect, deficiency of ceruloplasmin is found in Wilson's disease, Kayser-Fleischer rings] c. Depleted cellular hemosiderin levels [is incorrect, in hemochromatosis, the iron is continuously taken up and stored inside the cells in ferritin and hemosiderin. This leads to liver damage and serum ferritin increases] d. Reduced dietary iron uptake [is incorrect, it is a life-long uncontrolled iron uptake] e. Decreased transferrin in the blood [is incorrect, transferrin is the transport protein for ferric iron in blood. Not less but more ferric iron will be transported]
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Q#15	Vitamin B12 deficiency can be due to a reduced synthesis of intrinsic factor (IF) in the elderly. What is special about the intrinsic factor? e. is correct: It is needed for uptake of B12 into intestinal mucosal cells a. It is secreted by chief cells in the stomach [is incorrect, it is secreted by parietal cells] b. The IF-B12 complex is degraded by pancreatic proteases [is incorrect, the R-binder/B12 complex is degraded by pancreatic proteases, IF-B12 is formed in the duodenum and is not degraded] c. It transports vitamin B12 in the blood [is incorrect, transcobalamin transports vitamin B12 in blood] d. It is synthesized in intestinal mucosal cells [is incorrect, IF is synthesized by parietal cells of the stomach] e. correct
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Q#16	A 40-year old woman was admitted to the emergency room. She was in a highly agitated state, weeping and complaining about severe abdominal pain. After talking to the patient, the physician thought that his patient could suffer from Acute Intermittent Porphyrria and he ordered the rapid urine test analysis for porphobilinogen. After detection of porphobilinogen, the urine was further analyzed and showed also increased levels of ALA. What would be the most appropriate treatment? Administration of D. is correct: hemin. ALA synthase in the liver is normally inhibited by hemin. In AIP patients, the synthesis of heme in the liver is defective, and the synthesis continues to be stimulated. Administration of hemin leads to inhibition of the synthesis of ALA synthase in these patients. a. a blood transfusion [is incorrect, it is not a treatment for AIP, it would not inhibit ALA synthase] b. vitamin B6 [is incorrect, it would lead to more pyridoxal phosphate PLP, the coenzyme for ALA synthase, which is already over-stimulated in AIP] c. barbiturates [is incorrect, as barbiturates stimulate cytochrome P450 synthesis in the liver and with that synthesis of heme, which leads in these patients to dangerously high levels of ALA and porphobilinogen.] d. correct e. drugs against photosensitivity [is incorrect, the patient is not photosensitive, as AIP is the result of HMB synthase deficiency. Patients with a defect of heme synthesis after the formation of HMB (hydroxymethylbilane) are photosensitive due to "faulty" porphyrin rings and radical damage]
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Q#17	Which of the following injury markers in the blood is the most specific for liver damage due to ethanol abuse? Increased levels of E is correct: serum GGT increases after ethanol abuse a. Alanine aminotransferase [is incorrect, it a marker for general acute hepatocyte damage] b. Aspartate aminotransferase [is incorrect, it a marker for general hepatocyte damage] c. AST/ALT ratio smaller than 2 [is incorrect, AST/ALT larger than 2 is correct] - d Albumin [is incorrect, the levels are not increased, they can be decreased in cirrhosis] - e ☐ ☐ γ-glutamyl transpeptidase [is correct]
Q#18	Heme degradation involves many different cells like macrophages, hepatocytes, intestinal bacteria and kidney cells. A defect in this catabolism can lead to jaundice, which is due to high bilirubin levels in the blood (>3 mg/dL). What happens normally to bilirubin that is released from the reticuloendothelial system? Bilirubin B is correct: is taken into the liver and conjugated to bilirubin-diglucuronide. Normally, two UDP-glucuronic acids are used catalyzed by UDP-glucuronyl transferase (UDPGT) a. is transported in blood bound to ligandin [is incorrect, it is transported in blood bound (non-covalently) to albumin. After uptake into the liver, it is complexed to ligandin inside of the hepatocyte] b. correct - c can lead to kernicterus in an adult with jaundice [is incorrect, it can lead to kernicterus in infants, but not in the adult as it does not cross the adult blood-brain barrier.] - d is degraded in the fetus to urobilinogen [is incorrect, bilirubin is formed in the fetus and transfers the placenta. Fetal bilirubin is metabolized in the liver and intestines of the mother] - e is converted to urobilinogen in the hepatocyte [is incorrect, urobilinogen is formed in the intestines by bacteria from bilirubin.]
Q#19	Heme synthesis takes place mainly in the erythroid cells and in the hepatocytes. The synthesis starts and ends in mitochondria with cytosolic enzymes in between. What is the regulated enzyme of heme synthesis in hepatocytes? B is correct: ALA synthase . It is induced when heme is needed and it is inhibited when heme accumulates in the liver. a. ALA dehydratase [is incorrect, this enzyme is not the regulated enzyme, it forms porphobilinogen in cytosol] b. correct c. Uroporphyrinogen decarboxylase [is incorrect, this enzyme is not the regulated enzyme, it is deficient in patients with Porphyria Cutanea tarda] d. Ferrochelatase [is incorrect, ferrochelatase is not the regulated enzyme in the liver. It is the last enzyme in heme synthesis]. e. Uroporphyrinogen III synthase [is incorrect, it is not the regulated enzyme. The isozyme in bone marrow is deficient in Congenital Erythropoietic Porphyria]

Q#20	Jaundice can result from genetic defects or from acquired liver or biliary disease. One can group jaundice into prehepatic, hepatic, or posthepatic jaundice. What is characteristic for prehepatic jaundice ? E. is correct: It can occur following a massive lysis of RBC. This leads to high total bilirubin, which is the unconjugated bilirubin. a. Gallstones block the biliary tracts [is incorrect, gallstones lead to (complete) obstructive jaundice as gallstones reduce or prevent the release of bile from the liver into the duodenum] b. The levels of alkaline phosphatase are elevated in blood [is incorrect, elevated levels of sALP are found in obstructive jaundice. ALP is induced in irritated bile ducts and is backed up into the liver and it is released into the blood. In hemolytic jaundice this is not the case] c. The urine is of a reddish brown color due to bilirubin in urine. [is incorrect, the kidney can release only the water-soluble conjugated bilirubin into the urine. Hemolytic jaundice does not lead to conjugated (direct) bilirubin in the blood, the conjugated bilirubin is released into the bile] d. It leads to high levels of conjugated bilirubin in blood [is incorrect, hemolytic (prehepatic) jaundice leads to accumulation of unconjugated bilirubin in the blood. UDPGT is “overwhelmed” by high unconjugated bilirubin from hemolysis. The formed conjugated bilirubin is released into the bile and leads to urobilinogen] e. correct
Q#21	Ethanol metabolism affects the whole body. Most of the ethanol reaches the liver and is metabolized in hepatocytes to acetaldehyde and acetate. Which of the following changes of liver metabolism occurs at a high ethanol concentration in the liver? A is correct: Increased synthesis of triacylglycerols. High concentration of ethanol leads to a high NADH/NAD⁺ ratio in liver cytosol. The formation of glycerol-3-P from DHAP in cytosol is favored, and TAG synthesis is increased. b. Inhibition of ketone body formation during fasting [is incorrect, ketone body formation during fasting is stimulated] c. Decreased NADH/NAD⁺ ratio [is incorrect, the NADH/NAD⁺ ratio is increased in both cytosol and mitochondria] d. Inhibition of glycogen degradation [is incorrect, ethanol has no effect on glycogen degradation in the liver. It is gluconeogenesis that is inhibited at high NADH levels following ethanol consumption] e. Increased uptake of lactate for gluconeogenesis [is incorrect, lactate cannot be used for gluconeogenesis at high NADH levels, as lactate dehydrogenase needs NAD⁺ in order to form pyruvate. Pyruvate is used at high NADH level to form lactate which leaves into the blood. This can result in hypoglycemia and lactic acidemia.]
Q#22	The liver performs many pathways after a meal and during fasting. Which of the following pathways is not any more performed in the liver after two days of fasting? B. is correct: Glycogen degradation. After this time of fasting, the glycogen stores in the liver are mostly depleted. a. Gluconeogenesis [is incorrect, it is needed for release of glucose into the blood] b. correct

	c. Ketone body formation [is incorrect, ketone bodies are released during prolonged fasting] d. Urea cycle [is incorrect, the urea cycle has to be performed all the time] e. Fatty acid degradation [is incorrect, fatty acids are taken up from the blood and degraded]
Q#23	The effects of ethanol on liver metabolism depend on the actual ethanol concentration in the hepatocyte and on the fact whether chronic alcohol abuse is present. What happens in the liver after a high ethanol intake by a person who is not a chronic alcoholic? a. correct: Development of temporary hyperlipidemia. TAG synthesis is stimulated due to high NADH levels, and more VLDLs than normal are formed and released into the blood. This is a temporary hyperlipidemia, as after exclusion of ethanol uptake, the NADH levels in the liver normalize and the VLDL release is back to normal. b. Development of a fatty liver [is incorrect, fatty liver can be found in chronic alcoholics who have high acetaldehyde levels in the liver. This inhibits the release of VLDL and leads to fat accumulation inside the liver.] c. Development of liver cirrhosis [is incorrect, liver cirrhosis can result from fatty liver in chronic alcoholics when ethanol is not omitted] d. Activation of gluconeogenesis due to low cytosolic NADH levels [is incorrect as it is high NADH levels that inhibit usage of precursors for gluconeogenesis] e. Inhibition of ketone body formation [is incorrect, ketone body formation is activated.]
Q#24	Porphyrias are diseases related to heme synthesis and their clinical symptoms often depend on the accumulation of intermediates following the deficiency of specific enzymes. What is characteristic for Porphyria Cutanea Tarda? This porphyria C. is correct: is the result of the deficiency of uroporphyrinogen decarboxylase. This leads to accumulation of uroporphyrinogen III which is non-enzymatically changed to the red uroporphyrin III. Uroporphyrin accumulates in blood and urine. This leads to a reddish color of the urine. Patients are photosensitive and the skin of hands shows red blisters. a. leads to accumulation of protoporphyrin IX [is incorrect, accumulation of uroporphyrin occurs. Protoporphyrin IX accumulates in ferrochelatase deficiency]. b. leads to patients who are highly irritable but who are not photosensitive [is incorrect, this describes a patient with acute intermittent porphyria.] c. correct d. is the result of lead poisoning [is incorrect: lead poisoning leads to microcytic anemia and inhibition of ALA dehydratase and of ferrochelatase] e. is a very rare porphyria [is incorrect: it is a common porphyria]
Q#25	Jaundice is the result of elevated bilirubin levels in the blood. A mild jaundice in newborns or premature babies is common and it is referred to as D. is correct: physiological jaundice. This is due to a low activity of UDPGT in newborns. It is temporary until normal levels of UDPGT are synthesized in the babies. a. Crigler-Najjar syndrome I [is incorrect, Crigler-Najjar syndromes lead to severe jaundice. Infants with Type I can die due to kernicterus if untreated, and patients with Type II often need lifelong treatment. The blood

	bilirubin levels are very high (10-50 x normal) in Crigler-Najjar syndromes.] b. Obstructive jaundice [is incorrect, obstructive jaundice is mainly found in adults and can lead to severe jaundice] c. Gilbert's syndrome [is incorrect, it is a mild jaundice, but mostly found in adults (often young men but also in females) and it is triggered by infection or starvation. Elevation of bilirubin in blood is temporary] d. correct e. Arias syndrome [is incorrect, Arias syndrome is another name for Crigler-Najjar syndrome II. It is a rare genetic defect and leads to more severe jaundice than found in physiological jaundice]
Q#26	The liver performs many pathways that are also found in other cells. But the liver performs also pathways that are unique for hepatocytes. Which of the following synthetic pathways is unique for the liver? e. is correct: ketone body synthesis. The liver forms ketone bodies during fasting and releases 3-hydroxybutyrate and acetoacetate into the blood. Only the hepatocytes have remarkable levels of mitochondrial HMG CoA synthase. Other unique pathways of the liver are the urea cycle where arginase is found in the liver but not in the kidney, and primary bile acid synthesis from cholesterol and also formation of conjugated bile salts. a. Gluconeogenesis [is incorrect, it is also performed by the kidney] b. Glycogen synthesis [is incorrect, it is also performed by muscle] c. Fatty acid synthesis [is incorrect, it is also performed in other tissues like mammary gland and brain] d. Amino acid synthesis [is incorrect, it is performed in many tissues] e. correct
Q#27	You saw a picture of a patient with jaundice, and it was very clear how the yellow color showed up in the sclerae of his eyes due to accumulation of a specific intermediate of total heme degradation. Which of the following compounds is found in high levels in the blood and accumulates in skin and sclerae in patients with hepatocellular jaundice? B is correct: Bilirubin. The definition of jaundice is that the blood level of total bilirubin is high (>3 mg/dL). Bilirubin is a lipid that is transported in the blood complexed to albumin, but accumulates at abnormal high levels in the sclerae and skin (in infants also in the brain). Conjugated bilirubin is normally not found in the blood, but it can leak out of the liver into the blood in patients with intrahepatic (hepatocellular) or obstructive jaundice. Conjugated bilirubin is water soluble and can be released by the kidney into the urine, but it also can accumulate in the sclerae and the skin a. Urobilinogen [is incorrect, urobilinogen is colorless and it also does not accumulate in sclerae or skin, it is soluble in blood and it reaches the kidney.] b. correct c. Urobilin [is incorrect, urobilin is yellow but found in urine and not in the skin or sclerae. It is formed from urobilinogen in the kidney and by contact with air] d. Stercobilin [is incorrect, it is formed in the large intestine by bacteria from urobilinogen and is released in the

	feces, brown color] e. Biliverdin [is incorrect, biliverdin is reduced to bilirubin. High bilirubin levels lead to jaundice]
Q#28	Heme degradation is a process that involves several tissues and also bacteria. What is characteristic for heme degradation? It leads to A is correct: formation of carbon monoxide by heme oxygenase. CO has physiologic functions as signaling molecule and vasodilator. Heme oxygenase forms also biliverdin. b. conjugation of biliverdin with UDP-glucuronate in the liver [is incorrect, biliverdin is changed to bilirubin. Diglucuronyl-bilirubin is formed in the liver.] c. bilirubin that is covalently bound to albumin in blood. [is incorrect, bilirubin is complexed to albumin and in infants, it can be substituted by drugs (aspirin) which can lead to the dangerous kernicterus]. d. conjugated bilirubin that is formed in the gallbladder [is incorrect, it is formed in the hepatocyte. Conjugated bilirubin is released into the bile and can be stored in the gallbladder. Bile is concentrated in the gall bladder which can lead to yellow gallstones (cholesterol, common) or red gallstones (bilirubin)] e. brown urobilinogen in the large intestine [is incorrect, urobilinogen is colorless and leads to the brown stercobilin in the large intestine.]
Q#29	During fasting, the liver uses β-oxidation of fatty acids to generate eventually ATP and acetyl CoA, which are both needed for gluconeogenesis. Carbons of fatty acids cannot be used for gluconeogenesis, but acetyl CoA formed in β-oxidation is needed as A. is correct: absolute activator for pyruvate carboxylase. In this reaction, oxaloacetate is formed in mitochondria which will be changed to malate. Malate leaves mitochondria and oxaloacetate is formed in cytosol for PEP carboxykinase and gluconeogenesis. a. correct b. as absolute activator of PEP carboxykinase [is incorrect, it is needed as absolute activator for pyruvate carboxylase] c. substrate for acetyl CoA carboxylase [is incorrect, acetyl CoA formed during fatty acid degradation is formed inside of mitochondria, acetyl CoA carboxylase is in cytosol for fatty acid synthesis] d. substrate for cytosolic HMG CoA synthase [is incorrect, HMG CoA for cholesterol synthesis is formed in cytosol from three acetyl CoAs in cytosol] e. inhibitor of the carnitine shuttle [is incorrect, the carnitine shuttle is inhibited by cytosolic malonyl CoA]
Q#30	Enzyme induction is a tool to increase the actual amount of the enzyme. The hormones insulin and glucagon favor different pathways in the liver and induce different key enzymes. Which of the following enzymes is induced in the liver by insulin? C is correct: Glucokinase. Insulin induces in the liver glucokinase which is not product inhibited and can phosphorylate large amounts of glucose to glucose 6-P. a. Glucose 6-phosphatase [is incorrect, it is induced by glucagon] b. Hexokinase [is incorrect, insulin induces glucokinase in the liver] c. correct.

	d. PEP carboxykinase [is incorrect, it is induced by glucagon] e. Fructose 1,6-bisphosphatase [is incorrect, it is induced by glucagon]
Q#31	A 22-year old female student was brought by her friends to the emergency room at noon on New Years Day. She was sweating profusely, hyperventilating and scared by irregular heart beats. She was highly agitated and her breath smelled of alcohol. Her friends told the physician, that she had been drinking heavily during the whole night. She did not take medications and had slept for several hours. She suddenly woke up from unusual fast heart beats. Her lab results showed the following: Aspartate aminotransferase (AST)30 U/L (reference range 5-35) Alanine aminotransferase (ALT)50 U/L (reference range 5-56) Alkaline phosphatase (ALP)70 U/L (reference range 20-160) γ-glutamyl transpeptidase (GGT)157 U/L (reference range 8-78) Total bilirubin1.4 mg/dL (normal <1.4 mg/dL) Direct bilirubinnormal Albumin4.3 g/dL (reference range 3.5-5.5) Which of the following is most likely in the clinical scenario? B is correct: Lactic acidemia. Her high intake of ethanol resulted in the inability to use lactate by the liver for gluconeogenesis. At the same time, there is generation of more lactate from pyruvate at high NADH levels in liver cytosol, and more lactate is released into the blood. The other choices can be excluded by her blood data or general knowledge of ethanol metabolism. a. Liver cirrhosis [is incorrect, the normal levels of sALP together with normal levels of sALT , sAST, and albumin exclude liver cirrhosis] b. correct c. Hypoketonemia during fasting [is incorrect, high levels of NADH in mitochondria favor release of 3-hydroxybutyrate from the liver. Ethanol intoxication leads to a high NADH/NAD+ ratio] d. Obstructive jaundice [is incorrect, sALP and total bilirubin would be elevated, direct bilirubin would be increased] e. Hepatocellular jaundice [is incorrect, as in this patient sALT, sAST and total bilirubin are in reference range, direct bilirubin is absent]
Q#32	A 45-year old man complains about steatorrhea. The sclerae of his eyes are yellowish. Blood analysis shows high levels of direct bilirubin and increased alkaline phosphatase (ALP). The blood levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST), albumin and inactive blood clotting factors are normal. The dipstick method shows bilirubin in the urine. What is most likely the reason for her jaundice? a. correct: Gallstones. Gallstones lead to obstructive jaundice, characterized by high levels of sALP and by high levels of conjugated bilirubin (direct bilirubin) in the blood. This was also shown by bilirubin in the urine (dipstick). Hepatocytes are often not damaged and sALT and sAST are

	normal. As the conjugated bilirubin cannot be released via the bile into the intestine, it is pushed back into the liver and the water soluble conjugated bilirubin is released into the blood. b. Excessive hemolysis [is incorrect, hemolytic jaundice leads to elevated levels of unconjugated bilirubin, and one would not detect bilirubin in the urine. Steatorrhea is not common in hemolytic jaundice] c. Gilbert's syndrome [is incorrect, Gilbert's syndrome is common in adults, but it leads to unconjugated bilirubin in the blood as it is a deficiency of UDPGT (50%). The urine would not contain bilirubin.] d. Crigler-Najjar Type II [is incorrect, it would lead to high levels of unconjugated bilirubin. The urine would not contain bilirubin] e. Severe liver damage [is incorrect, sALT and sAST and albumin and blood clotting factor levels are normal. The problem is related to the bile ducts indicated by elevated levels of sALP.]
Q#33	Ethanol metabolism in the liver starts with alcohol dehydrogenase. This enzyme has a large variety of activity dependent on genetic background. What is in general characteristic for alcohol dehydrogenase? This enzyme C is correct: forms acetaldehyde in cytosol. Acetaldehyde can then enter mitochondria or stay in the cytosol for further metabolism by the respective acetaldehyde dehydrogenases a. forms NADH in mitochondria [is incorrect, alcohol dehydrogenase forms NADH in the cytosol] b. has a larger Km for ethanol than CYP 2E1 [is incorrect, the Km for ethanol is smaller for alcohol dehydrogenase than for CYP 2E1] c. correct d. forms acetate in mitochondria [is incorrect, acetate is formed from acetaldehyde by acetaldehyde dehydrogenase] e. is part of MEOS, the microsomal ethanol oxidizing system [is incorrect. MEOS (includes CYP 2E1) also forms acetaldehyde from ethanol but it is a different enzyme system, other than alcohol dehydrogenase. It has a larger Km for ethanol, but can be induced by ethanol. During its action, radicals are formed that are normally immediately scavenged]
Q#34	Hemoglobin in mature RBC was synthesized in erythroid cells. This included heme synthesis while mitochondria were still available. What is special of heme synthesis in erythroid cells? It C is correct: leads to synthesis of globin chains. In erythroid cells, heme synthesis leads to as much as possible synthesis of hemoglobin, which will be needed in the mature RBC. It is a one-time synthesis, as mature RBCs do not contain mitochondria any more, whereas erythroid cells do. a. starts and ends in the cytosol [is incorrect, it starts and ends in mitochondria] b. is inhibited by hemin [is incorrect, in erythroid cells, heme synthesis is limited by availability of iron (ALAS2 is inhibited at low iron levels). In the liver, ALAS1 is inhibited by hemin.] c. correct d. is not inhibited by lead [is incorrect, lead inhibition inhibits ALA dehydratase and also ferrochelatase in all cells that synthesize heme. The anemia shows up related to erythroid cells] e. leads to heme catalyzed by ferrochelatase

	[is incorrect, heme contains ferric iron, whereas ferrochelatase inserts ferrous iron into protoporphyrin IX and forms heme]
Q#35	MCAD deficiency is the most common genetic defect in fatty acid metabolism. The acute symptoms are similar to Reye syndrome and SIDS. What is the most specific observation found in patients with MCAD deficiency which is not found in other diseases? C. is correct: dicarboxylic acids in urine. The accumulated medium-chain fatty acids are changed by ω-oxidation and their respective dicarboxylic acids are found in blood and urine. (We find also fatty acyl-carnitines in the blood) a. Ketoacidosis [is incorrect, ketoacidosis cannot be found in patients with MCAD deficiency, they are hypoketotic] b. Hypoglycemia [is incorrect, it is found in MCAD deficiency patients but it is not specific. It can be found in many diseases, like for example Von Gierke's disease] c. correct. d. Inhibition of CPT II [is incorrect, this is not found in MCAD deficiency. It would inhibit the transport of long-chain acyl groups into mitochondria] e. Hyperglycemia and hypoketonemia [is incorrect, MCAD deficiency leads to hypoglycemia and hypoketonemia]
Q#36	A 40-year old woman is admitted to the emergency room. She fell down the stairs and still smells of alcohol. The neighbor who alerted the emergency team pointed out that the patient severely abused alcohol for several years. The attending physician decides to administer vitamins as part of her treatment of chronic alcoholism. The cofactor pyridoxal phosphate is formed from vitamin B6 and is needed as coenzyme for a. Is correct: GABA formation from glutamate. The decarboxylation of the α-carboxyl group of glutamate needs PLP as coenzyme. As C#1 is lost, the molecule is renamed using the former γ-carboxyl-group as C#1 which leads to γ-aminobutyric acid (GABA) a. correct b. methionine formation from homocysteine [is incorrect, it needs methyl-THF and B12] c. deoxyribonucleotide formation from ribonucleotides [is incorrect , it needs 2 reduced thioredoxins] d. purine nucleotide degradation [is incorrect, it needs molybdenum for xanthine oxidase] e. degradation of catecholamines [is incorrect, it needs SAM for COMT]
Q#37	During your clinical rotations, you are asked to take the medical history of a 3-year old boy. You come to the conclusion that this patient might suffer from Tay-Sachs disease. The reasoning for this genetic defect is the origin of his parents, specific onion-shell-like accumulations in his cells and that he shows b. is correct: a cherry-red macula. [This is seen in patients with Tay-Sachs disease and Niemann-Pick disease. Tay-Sachs disease leads to onion-shell-like accumulations of GM2, Niemann-Pick

	disease leads to foamy appearing cells containing sphingomyelin.] a. hypopigmentation [is incorrect, that is found in other diseases like tyrosinase deficiency] b. correct c. a blue sclera [is incorrect, this can be an indication for osteogenesis imperfecta] d. signs of self-mutilation [is incorrect, this is found in patients with Lesch-Nyhan Syndrome] e. a dislocated lens [is incorrect, this can be found in patients with Marfan Syndrome]
Q#38	Which of the following compounds is needed for dopa decarboxylase? c. Is correct: PLP is needed for dopa decarboxylase which forms dopamine. In general, decarboxylations of the α-carboxyl-group of amino acids needs PLP. Other examples are histidine decarboxylase (histamine), glutamate decarboxylase (GABA) and hydroxytryptophan decarboxylase (serotonin). a. copper and vitamin C [is incorrect, it is needed for dopamine hydroxylase] b. tetrahydrofolate [is incorrect, it is needed in one-carbon transfer] c. correct d. thiamine pyrophosphate [is incorrect, it is needed for PDH, α-ketoglutarate DH, transketolase] e. SAM [is incorrect, it is a major methylation compound]
Q#39	A 2-year-old girl is treated with enzyme replacement therapy by intravenous infusion in the hospital. She is short of stature, has coarse facial features and shows macrocephaly and hepatosplenomegaly. Her joint mobility is severely restricted. The modified enzyme that is used in her enzyme replacement therapy is iduronidase, which is given to patients with E is correct: Hurler Syndrome a. Gaucher Disease [is incorrect, the deficient enzyme is glucocerebrosidase] b. Hunter Syndrome [is incorrect, it is an X-linked deficiency of iduronate sulfatase] c. Tay-Sachs Disease [is incorrect, the deficient enzyme is hexosaminidase A] d. Niemann-Pick Disease [is incorrect, the deficient enzyme is sphingomyelinase] e. correct
Q#40	A 14-year old boy plans to travel with his parents to Egypt and he comes to the clinic for a routine check-up. On this occasion, he tells the physician that he has trouble sweating. On physical examination, the boy is found to have symptoms that are consistent with Fabry's disease, especially as he has dark reddish skin lesions with bathing trunk distribution. What would most likely pathologically accumulate

	in his vascular endothelium? b. is correct: globosides. The deficient enzyme is α-galactosidase which leads to the trihexose-ceramide Gal-Gal-Glc-Cer (which is a globoside) a. galactocerebrosides [is incorrect, they accumulate in Krabbe Disease] b. correct c. glucocerebrosides [is incorrect, they accumulate in Gaucher Disease] d. sphingomyelin [is incorrect, it accumulates in Niemann-Pick Disease] e. ganglioside GM2 [is incorrect, they accumulate in Tay-Sachs Disease]
Q#41	A 15 year old girl has been admitted to the Burns unit. She has had severe burns when there was an accidental fire in the house. She has been on fluid and electrolytes on admission and her blood pressure has been stabilized. On the third day in the hospital, which of the following metabolic changes would you expect? D is correct: the change would be an increased rate of muscle proteolysis. a. A decrease in circulating cortisol [is incorrect, cortisol level would be increased] b. An increase in ketone body production [is incorrect, ketone body production is not increased in patients with trauma] c. Inhibition of hormone sensitive lipase in adipose tissue [is incorrect, hormone sensitive lipase in fat cells will be activated] d. correct e. A decrease in the metabolic rate [is incorrect, the metabolic rate will be increased] (Recapitulate the hormonal and metabolic changes in the ebb phase, flow phase and anabolic phase following trauma)
Q#42	An obese subject has been diagnosed to have the metabolic syndrome (syndrome X). Which of the following features would be present in the subject? C is correct: increased blood triglyceride levels would be present in this patient. a. Pear shaped distribution of body fat [is incorrect, it would be apple shaped distribution] b. Low levels of circulating insulin [is incorrect, high serum levels of insulin are expected] c. correct d. BMI of 24.6 [is incorrect, the BMI would be higher than normal] e. Increased activity of lipoprotein lipase [is incorrect, metabolic syndrome leads to reduced activity of LPL due to insulin resistance] (Identify the clinical and biochemical features of obesity and metabolic syndrome)

Q#43	A child is suspected as having kwashiorkor. Which of the following parameters would be diagnostic of kwashiorkor? B. is correct: low serum albumin levels are found in kwashiorkor. a. BMI greater than 25 [is incorrect, the BMI would be lower than 18.5] b. correct c. Increased waist circumference [is incorrect, this is not characteristic as it is found also in obese patients] d. Increased blood urea levels [is incorrect, they are observed in patients with renal disease but not in kwashiorkor] e. Increased skin fold thickness [is incorrect, skin fold are loose and less thick than normal due to loss of adipose tissue] (Know the differentiating features of marasmus and kwashiorkor)
Q#44	A 20 year old female model is practically starving for the past 15 days to appear for a beauty pageant. She has lost 10 pounds over the past 10 days. She has lost consciousness and has been brought to the ER. Regarding the metabolic processes at this time (compared to the postabsorptive phase), the TRUE statement is E is correct: the production of 3-hydroxybutyrate would be increased a. Adipose tissue hormone sensitive lipase would be inhibited [is incorrect, hormone-sensitive lipase in fat cells is activated] b. The insulin glucagon ratio would be increased [is incorrect, the insulin glucagon ratio would be decreased] c. The major source of gluconeogenesis would be lactate [is incorrect, the major source for gluconeogenesis would be amino acids] d. The metabolic rate would be increased [is incorrect, the metabolic rate would be decreased] e. correct (Identify the hormonal and metabolic adaptations in fed state, fasting state and on starvation)
Q#45	A 60 year old man on routine check-up was found to have an elevated serum cholesterol level. Which would be the most appropriate nutritional advice? C is correct: increase the intake of ω3 fatty acids in the diet a. Increase the saturated fatty acid content in the diet [is incorrect, saturated fatty acid content shall be low] b. Decrease the content of arachidonic acid in the diet [is incorrect, this ω6 fatty acid is beneficial] c. Correct d. Decrease the content of oleic acid in the diet [is incorrect, an increase of oleic acid is beneficial] e. Decrease the dietary fiber content [is incorrect, increased dietary fiber is beneficial]

	(Identify the importance of dietary lipids in relation to dyslipidemia)
Q#46	A 17 year old boy has been admitted for the evaluation of weight loss over the past few weeks and the complaints of increased frequency of urination at night. Fasting blood glucose levels were elevated on two consecutive occasions. A presumptive diagnosis of type I diabetes mellitus has been considered. Which of the following metabolic features would support the diagnosis? C is correct: low levels of circulating insulin are found. a. Presence of insulin resistance [is incorrect, this is found in type II diabetes mellitus] b. Presence of high circulating proinsulin levels [is incorrect, this would indicate increased endogenous insulin production] c. correct d. Increased circulating leptin levels [is incorrect, type I diabetes patients are generally not obese] e. Low levels of glycated hemoglobin [is incorrect, HbA_{1c} levels would be increased due to hyperglycemia] (Differentiate between type I and type II diabetes mellitus, identify the significance of all the lab tests mentioned above)
Q#47	A 40 year old man has been brought to the ER in the unconscious state. He has a daughter diagnosed to have type I diabetes mellitus and on maintenance insulin therapy. The physician suspects that the man has intentionally induced hypoglycemia by insulin injections. Which of the following lab results would indicate that the hypoglycemia is factitious? C is correct: decreased C peptide levels indicate insulin injection. This man has factitious hypoglycemia as he injected himself with insulin without being diabetic. a. Increased circulating insulin levels [is incorrect, insulin levels per se do not indicate factitious hypoglycemia. They can be due to insulinoma] b. Increased proinsulin levels [is incorrect, increased proinsulin levels indicate endogenous insulin synthesis] c. correct d. Presence of sulfonylurea in circulation [is incorrect, sulfonylurea is a drug that simulates insulin synthesis and elevates C-peptide levels. This drug was not available for this man, whereas insulin was] e. Increased ketone bodies in urine [is incorrect, ketone body levels would be reduced in a patient following insulin injection] (Identify the cause of hypoglycemia using the history and lab results, analyze the significance of each of the lab results)
Q#48	A 30-year-olds woman is in the hospital and her serum analysis shows elevated levels of α-fetoprotein. This protein

	C is correct: is a serum marker for liver cancer a. Is part of the α_2-globulin fraction [is incorrect, it is part of the α_1-globulin fraction] b. Is synthesized by activated B-cells [is incorrect, it is synthesized by the liver] c. correct d. Is increased in patients with multiple myeloma [is incorrect, multiple myeloma leads to a sharp peak of a single immunoglobulin] e. Is one of the largest serum proteins [is incorrect, it is α_2-macroglobulin that is one of the largest serum proteins]
Q#49	A war prisoner is on a hunger strike and reaches 40 days of starvation. Which of the following fuel sources is mostly used in his brain at this time? B is correct: after 40 days of starvation the brain uses mostly 3-hydroxybutyrate a. Free fatty acids [is incorrect, fatty acids cannot pass the blood-brain barrier, exception are dietary essential fatty acids] b. correct c. Acetoacetate [is incorrect, acetoacetate is used but less than 3-hydroxybutyrate] d. Glucose [is incorrect, glucose is still used but the amount is reduced] e. Amino acids [is incorrect, amino acids are not the fuel source mostly used after 40 days]
Q#50	Which of the following serum levels could best indicate chronic liver disease/cirrhosis? A is correct: decreased albumin can indicate liver cirrhosis a. correct b. Increased ALT [is incorrect, ALT increase signals acute liver damage and ALT levels decline during cirrhosis] c. Decreased ALP [is incorrect, ALP levels can even slightly be elevated in cirrhosis coming from the intra-hepatic bile ducts that are damaged] d. Decreased γ-globulins [is incorrect, γ-globulins increase in serum as response to cirrhosis] e. Increased α_2-macroglobulin [is incorrect, α_2-macroglobulin is not increased during cirrhosis, it is increased in nephrotic syndrome]